

A ‘c0’ Counterexample to Reverse One-Variable Steinhaus Decoupling

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Status

This packet gives a counterexample to removing the K -convexity hypothesis from the reverse one-variable Steinhaus decoupling estimate in Carando–Marceca–Sevilla-Peris [1]. It shows that the estimate fails for the non- K -convex space c_0 .

Source Question

The source paper proves that, if X is K -convex and $p > 1$, then for every m -homogeneous tetrahedral polynomial $P : \mathbb{C}^n \rightarrow X$, with associated symmetric m -linear map M ,

$$\sqrt{m}(\mathbb{E}\|M(w', w, \dots, w)\|^p)^{1/p} \leq C_{p,X}(\mathbb{E}\|P(w)\|^p)^{1/p},$$

where w, w' are independent Steinhaus vectors. The authors then ask whether the K -convexity assumption is necessary.

$$\sqrt{m}(\mathbb{E}\|M(w', w, \dots, w)\|^p)^{1/p} \geq \frac{1}{2}\sqrt{m}C_{p,X}(\mathbb{E}\|P(w)\|^p)^{1/p}.$$

Remark 4.3. The key fact in the proof of Proposition 4.1 is that for gaussian variables $a\gamma + b\gamma' \sim \sqrt{|a|^2 + |b|^2}\gamma''$ for every $a, b \in \mathbb{C}$. This is no longer true for other random variables such as Steinhaus variables. To obtain a similar behavior we take advantage of the geometry in $\mathbb{C}$ and use the variables w and $i\varepsilon w$ which always stay orthogonal to each other. Notice that in this case for $a, b \in \mathbb{R}$ we have that $aw + bi\varepsilon w \sim \sqrt{a^2 + b^2}w$. This explains the K -convexity requirement which arises from estimating the norm of homogeneous projections involving Rademacher variables we are forced to use in order to maintain the orthogonality between w and $i\varepsilon w$. We do not know if the K -convexity assumption is necessary. It is worth mentioning that one can retrieve the full-decoupling inequality (7) for Steinhaus variables by starting with the previous result and proceeding inductively. Therefore, weakening the K -convexity condition to assuming only finite cotype would provide a new proof of Corollary 2.1 for Steinhaus variables and possibly lead to better estimates.

We begin by showing a slightly more general fact than (33) which will later lead to an interesting application.

Lemma 4.4. Let $P : \mathbb{C}^n \rightarrow X$ be an m -homogeneous tetrahedral polynomial and fix $\lambda \in \mathbb{C}^n$. Then

Figure 1: Source evidence from arXiv:2012.15293, PDF page 24, Remark 4.3.

Counterexample

Theorem 1. For $X = c_0$, and for every $p \geq 1$, there is no constant C_p such that

$$\sqrt{m}(\mathbb{E}\|M(w', w, \dots, w)\|_{c_0}^p)^{1/p} \leq C_p(\mathbb{E}\|P(w)\|_{c_0}^p)^{1/p}$$

for all m, n and all m -homogeneous tetrahedral polynomials $P : \mathbb{C}^n \rightarrow c_0$.

Proof. Fix m and n . Let $\mathcal{P}_m[n]$ denote the family of m -element subsets of $[n] = \{1, \dots, n\}$. We identify the finite-dimensional space $\ell_\infty(\mathcal{P}_m[n])$ isometrically with a coordinate subspace of c_0 . Let $(e_A)_{A \in \mathcal{P}_m[n]}$ be its canonical basis and define

$$P(z) = \sum_{A \in \mathcal{P}_m[n]} e_A z_A, \quad z_A = \prod_{j \in A} z_j.$$

For every Steinhaus vector $w \in \mathbb{T}^n$,

$$\|P(w)\|_\infty = \sup_{A \in \mathcal{P}_m[n]} |w_A| = 1.$$

Hence $(\mathbb{E}\|P(w)\|_\infty^p)^{1/p} = 1$.

Let M be the symmetric m -linear map associated to P . For a fixed coordinate A , symmetrization gives

$$M_A(z^{(1)}, \dots, z^{(m)}) = \frac{1}{m!} \sum_{\sigma \in S_m} \prod_{r=1}^m z_{a_{\sigma(r)}}^{(r)}, \quad A = \{a_1, \dots, a_m\}.$$

Substituting $(z^{(1)}, z^{(2)}, \dots, z^{(m)}) = (w', w, \dots, w)$ yields

$$M_A(w', w, \dots, w) = \frac{1}{m} \sum_{j \in A} w'_j \prod_{\substack{i \in A \\ i \neq j}} w_i.$$

Since $|w_i| = 1$, if $u_j = w'_j/w_j$, then

$$|M_A(w', w, \dots, w)| = \frac{1}{m} \left| \sum_{j \in A} u_j \right|.$$

The random variables u_j are independent Steinhaus variables.

Let

$$\Gamma = \{e^{i\theta} : |\theta| \leq \pi/3\}.$$

Then $\mathbb{P}(u_j \in \Gamma) = 1/3$, and if $u_j \in \Gamma$ then $\operatorname{Re} u_j \geq 1/2$. Choose $n = n(m)$ large enough that, with probability at least $1/2$, at least m of the variables $u_1, \dots, u_n$ lie in Γ . This is possible because the number of such variables has binomial distribution $\operatorname{Bin}(n, 1/3)$, whose mean tends to infinity with n .

On this event, choose an m -set A among those good indices. Then

$$\left| \sum_{j \in A} u_j \right| \geq \operatorname{Re} \sum_{j \in A} u_j \geq \frac{m}{2},$$

and therefore $\|M(w', w, \dots, w)\|_\infty \geq 1/2$. Consequently

$$(\mathbb{E}\|M(w', w, \dots, w)\|_\infty^p)^{1/p} \geq \frac{1}{2} 2^{-1/p}.$$

If the displayed reverse one-variable decoupling estimate held in c_0 with a constant C_p independent of m and n , applying it to these polynomials would give

$$2^{-1-1/p} \sqrt{m} \leq C_p$$

for every m , a contradiction. Thus no such estimate can hold for c_0 . $\square$

Scope

The counterexample shows that the K -convexity assumption cannot simply be discarded. It does not decide the finer finite-cotype weakening discussed in the source remark, since c_0 has trivial cotype.

Verification Notes

No computer verification is needed. The proof has two checks: the symmetric multilinear coordinate formula and the elementary binomial arc-counting argument for independent Steinhaus ratios w'_j/w_j .

Novelty and Search Bounds

Before packaging, the run's cheap indexes were searched for arXiv:2012.15293, “one-variable decoupling”, “K-convexity”, “Steinhaus”, “ c_0 ”, and “ell_infty”. No prior packet or attempt for this exact target was found.

Human Review Recommendation

Review as a short counterexample packet. The only substantive mathematical point is that the supremum over all m -sets A lets a large number of variables expose an almost-aligned Steinhaus block, while the original polynomial has constant supremum norm 1.

References

- [1] D. Carando, F. Marceca, and P. Sevilla-Peris, *Decoupling inequalities with exponential constants*, arXiv:2012.15293.